Kernix 0.6.0 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

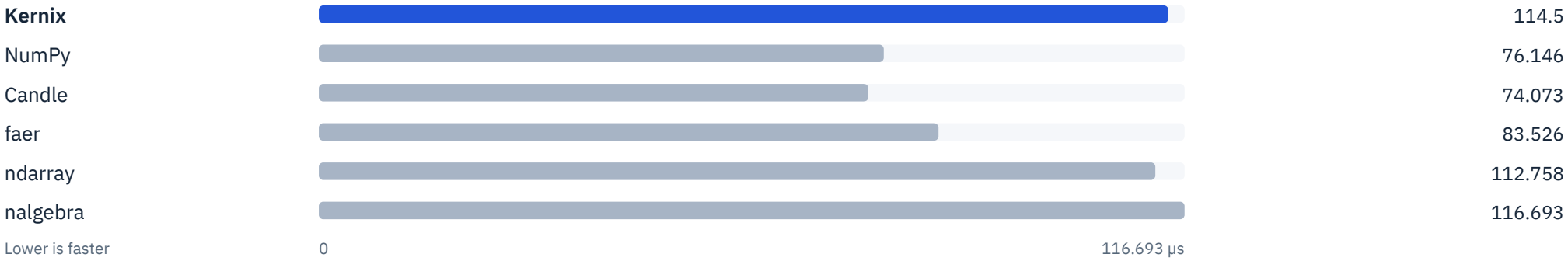
10 workloads

43 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

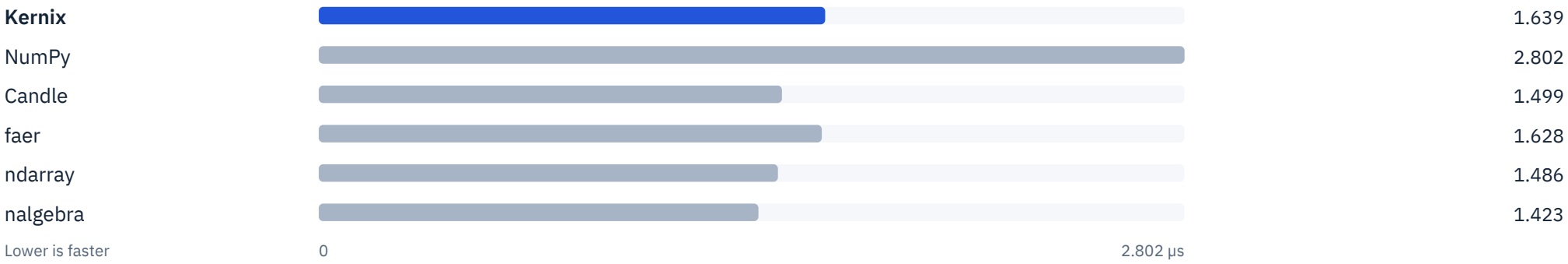
Kernix 0.6.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Batched multiplication

Per-call median

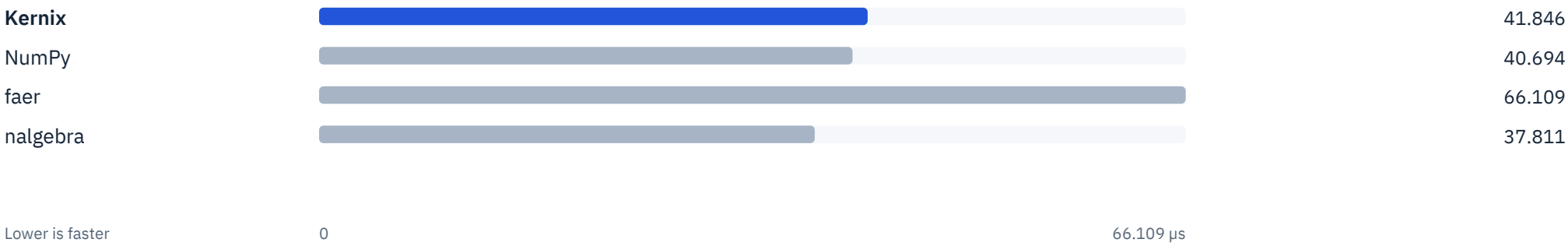
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches

Kernix 0.6.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

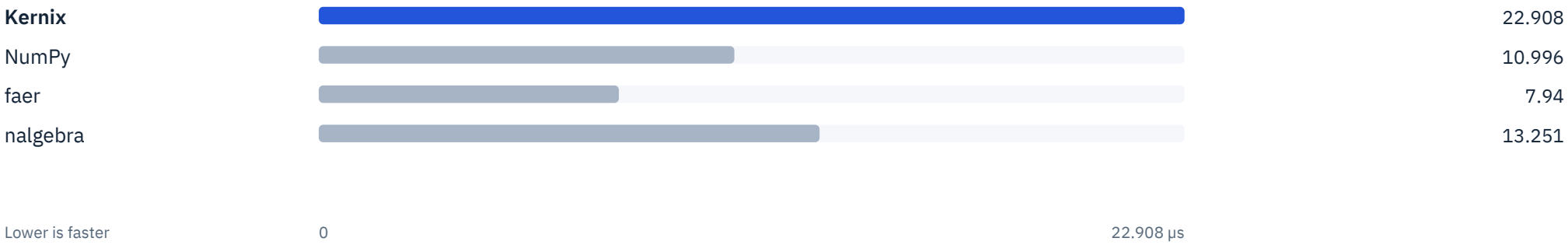
256 × 64 · f64 · 1 worker · allocated output

Kernix 0.6.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



Kernix 0.6.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Record + backward

Per-call median

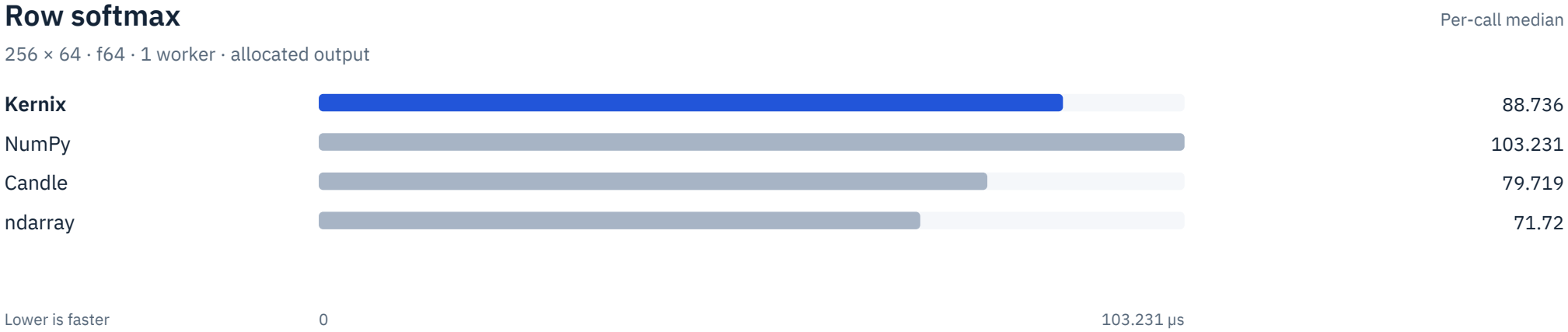
sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product

Kernix 0.6.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

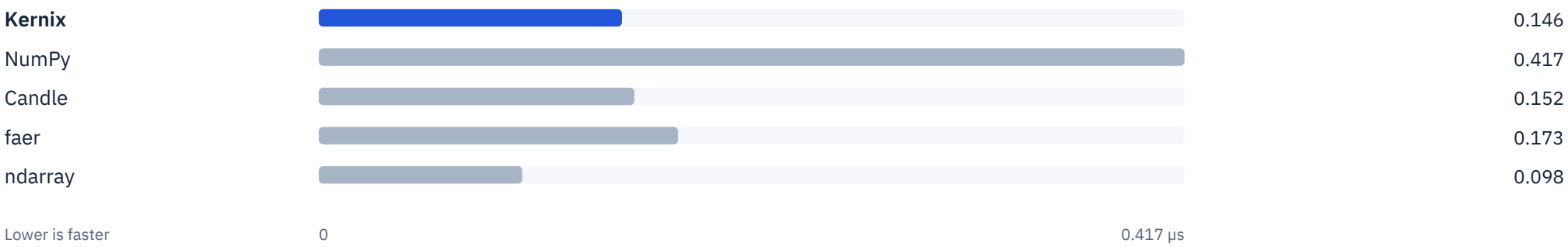


Kernix 0.6.0 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output



Create ones

64 × 64 · f32 · 1 worker · allocated output

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1639	1.639
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1486	1.486
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1423	1.423
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1628	1.628
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1499	1.499
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2802	2.802
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	114500	114.5
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	112758	112.758
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	116693	116.693
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	83526	83.526

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	74073	74.073
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	76146	76.146
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	280496	280.496
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	2120	2.12
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	3250	3.25
Thin QR / Kernix 96 × 32 · f64	41846	41.846
Thin QR / nalgebra 96 × 32 · f64	37811	37.811
Thin QR / faer 96 × 32 · f64	66109	66.109
Thin QR / NumPy 96 × 32 · f64	40694	40.694
Thin QR / Kernix 256 × 64 · f64	503390.99999999994	503.39099999999994

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / nalgebra 256 × 64 · f64	377013	377.013
Thin QR / faer 256 × 64 · f64	406409	406.409
Thin QR / NumPy 256 × 64 · f64	351733	351.733
Cholesky factorization / Kernix 64 × 64 · f64	22908	22.908
Cholesky factorization / nalgebra 64 × 64 · f64	13251	13.251
Cholesky factorization / faer 64 × 64 · f64	7940	7.94
Cholesky factorization / NumPy 64 × 64 · f64	10996	10.996
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3084	3.084
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	4206	4.206
Row softmax / Kernix 256 × 64 · f64	88736	88.736

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Row softmax / ndarray 256 × 64 · f64	71720	71.72
Row softmax / Candle 256 × 64 · f64	79719	79.719
Row softmax / NumPy 256 × 64 · f64	103231	103.231
Create zeros / Kernix 64 × 64 · f32	146	0.146
Create zeros / ndarray 64 × 64 · f32	98	0.098
Create zeros / faer 64 × 64 · f32	173	0.173
Create zeros / Candle 64 × 64 · f32	152	0.152
Create zeros / NumPy 64 × 64 · f32	417	0.417
Create ones / Kernix 64 × 64 · f32	203	0.203
Create ones / ndarray 64 × 64 · f32	155	0.155

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create ones / faer 64 × 64 · f32	190	0.19
Create ones / Candle 64 × 64 · f32	218	0.218
Create ones / NumPy 64 × 64 · f32	1317	1.317